

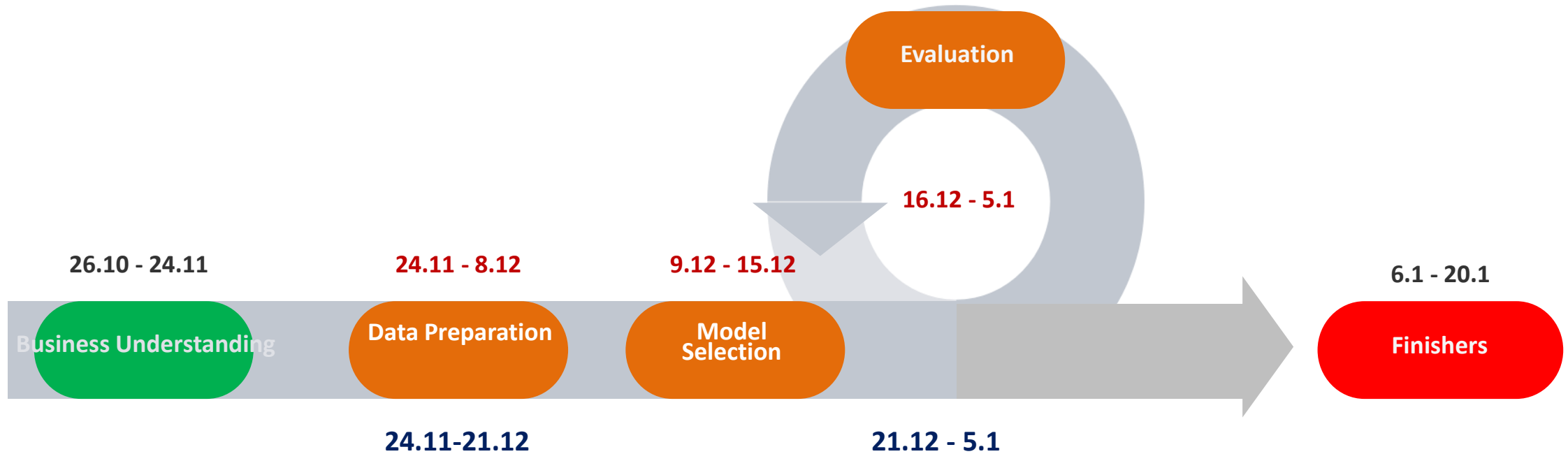


Check Point 1

Stocks Prediction Project

Workshop in Data Science
Team 003 - Itay Mutzafi, Moran Zaks , Shaked Schnarch

Project Status



Updated dates

Planned dates

Project Status - Finished Missions



Data Preparation

- Init Repo
- Fetch Data
- EDA
- Project Scope – 4 stocks

Evaluation

- Evaluation Metrics
- Evaluation Functions

Project Status – What's Next



Data Preparation

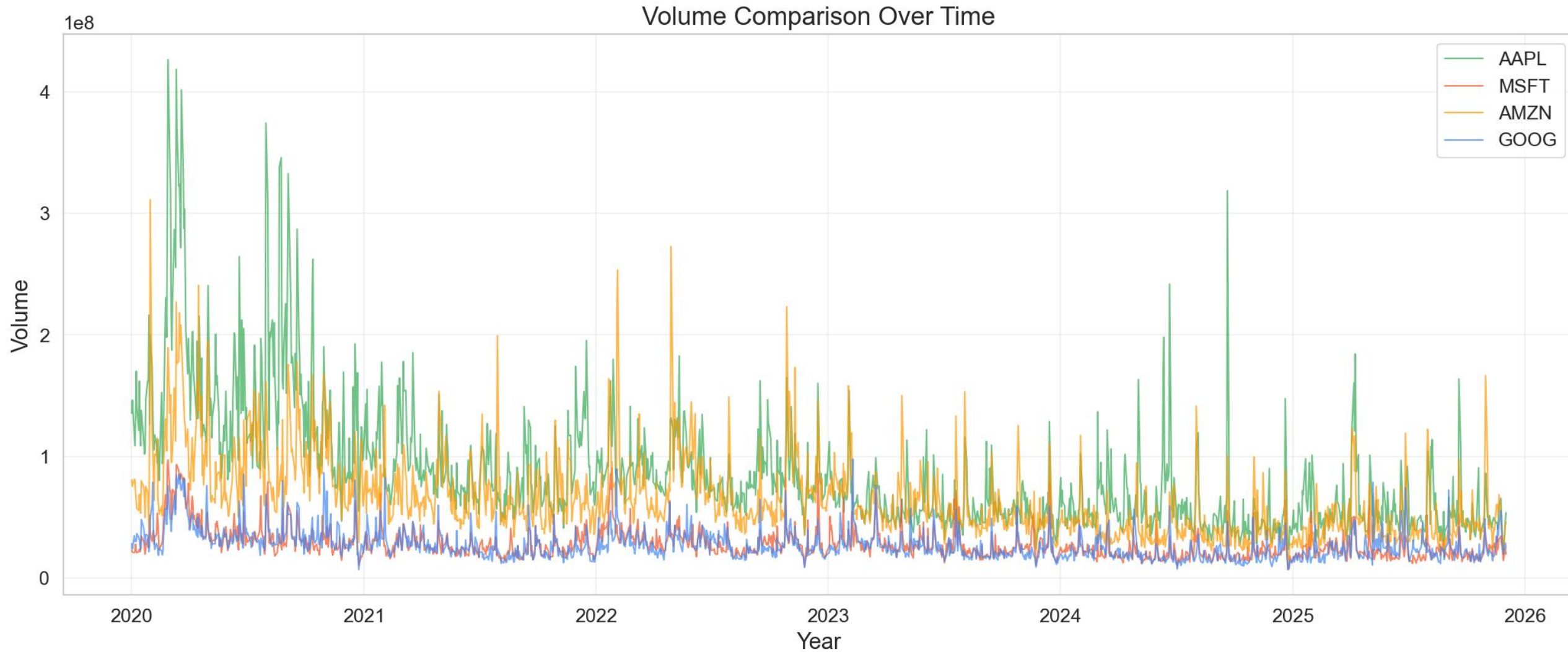
- Data Cleaning
- Feature Engineering (Yahoo!) – Days to report
- Sentiment Analysis (News)
- Test – Train split

Model Selection

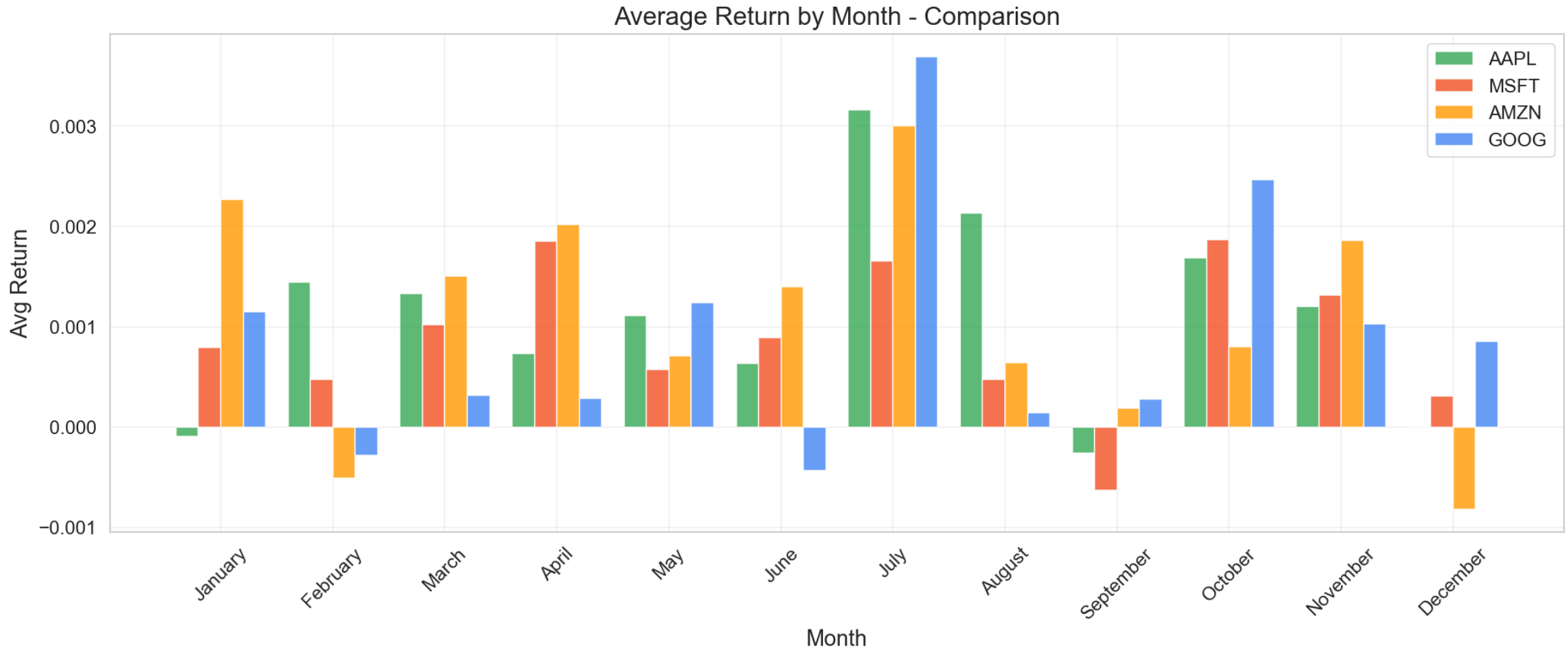
- Run Random Forest
- Run Regression Model

80% has been done, new deadline – 21.12
2 weeks of delay

Yahoo! – EDA General



Return - Seasonality



Seasonality – Significant Test

AMZN:



SEASONALITY TEST RESULTS

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Baseline period: Friday

Seasonality detected: NO ✗



Global significance tests

	Statistic	Value
0	ANOVA F	0.4522
1	ANOVA p-value	0.7708
2	Kruskal H	5.524
3	Kruskal p-value	0.2377



Effects relative to baseline (Friday)

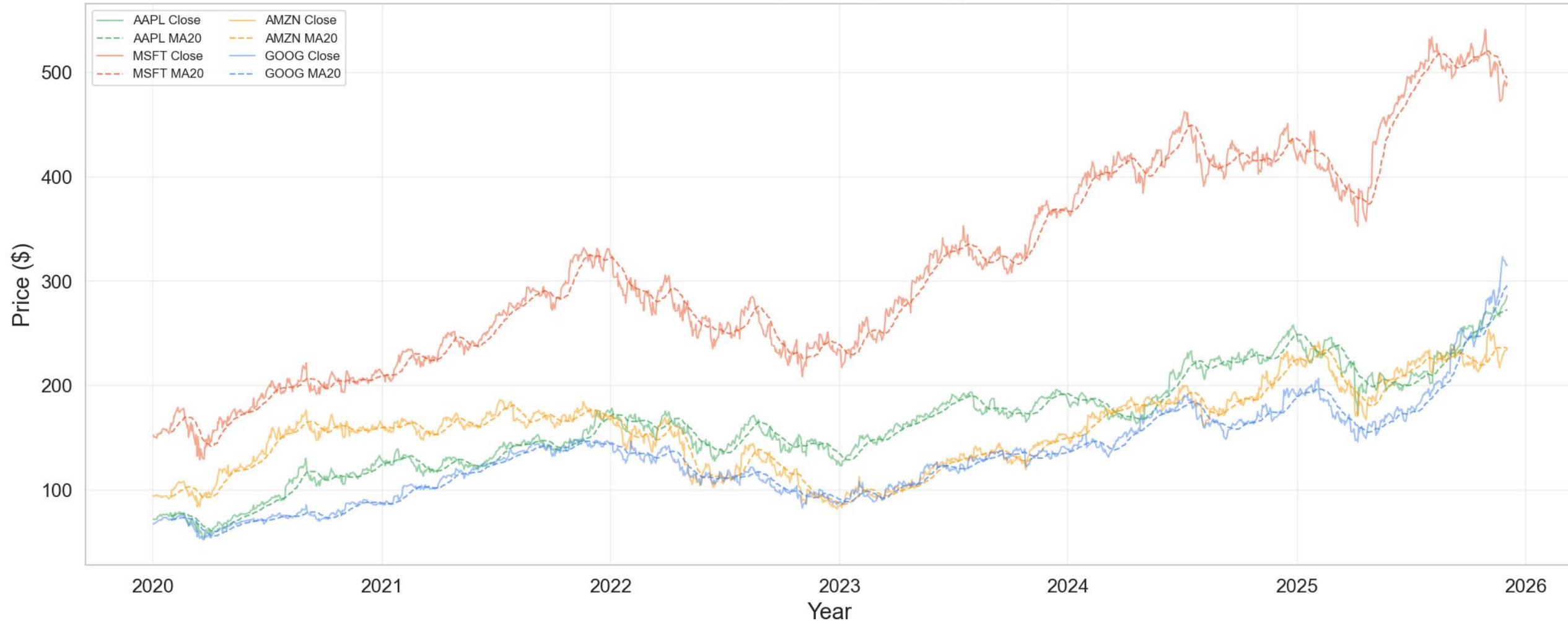
	Period	Coefficient	p_value	Significant	Direction_vs_baseline
0	C(Day)[T.Monday]	0.00007	0.9464	No	Higher
1	C(Day)[T.Thursday]	0.00081	0.4324	No	Higher
2	C(Day)[T.Tuesday]	0.00114	0.2662	No	Higher
3	C(Day)[T.Wednesday]	0.00070	0.4976	No	Higher



SEASONALITY TEST RESULTS

EDA - Engineered Features

Close Price and 20-day Moving Average - Comparison



Methodology: Formulation, Features & Evaluation

